

Andes Virus Outbreak

MV Hondius Cruise Ship · 2026

27 May 2026

Table of contents

Background	1
Methods	1
Results	2
Timeline of Exposure Windows	2
Genomic Analysis	3
Model Diagnostics	4
Transmission Tree	5
Ancestry Uncertainty	6
Dates of Infection	7
Imports and Unobserved Transmission	8
Reproduction Number	9
Note	9
Data	9

Outbreak summary · 27 May 2026

13 cases in analysis (11 confirmed · 2 probable) · 3 deaths · 03 Apr – 25 May 2026

Background

Andes virus (ANDV) is the only hantavirus documented to transmit person-to-person, in addition to the usual rodent-to-human route. The 2026 outbreak is linked to the **MV Hondius** expedition cruise ship. Cases first appeared in early April 2026; by the time infection was confirmed, passengers and crew had dispersed across Europe, Asia, and the Americas. Whole-genome sequencing confirms ANDV as the causative agent.

Methods

Transmission trees were inferred with **outbreaker2**, integrating symptom onset dates and genome sequences. Serial interval was modelled as Gamma(mean = 18 d, CV = 1/3). MCMC: 100,000 iterations, thinned every 50 steps, 10% burn-in. L-segment sequences were retrieved from [Pathoplexus](#) for the 4 of 13 cases with deposited accessions. Remaining cases contribute temporal data only. Because near-zero genetic variation renders the per-site mutation rate (μ) non-identifiable from the

data, μ was fixed at 1.7×10^{-5} substitutions/site/generation, derived from the published ANDV substitution rate of $\sim 3.5 \times 10^{-4}$ subs/site/year and an 18-day generation interval.

Results

Timeline of Exposure Windows

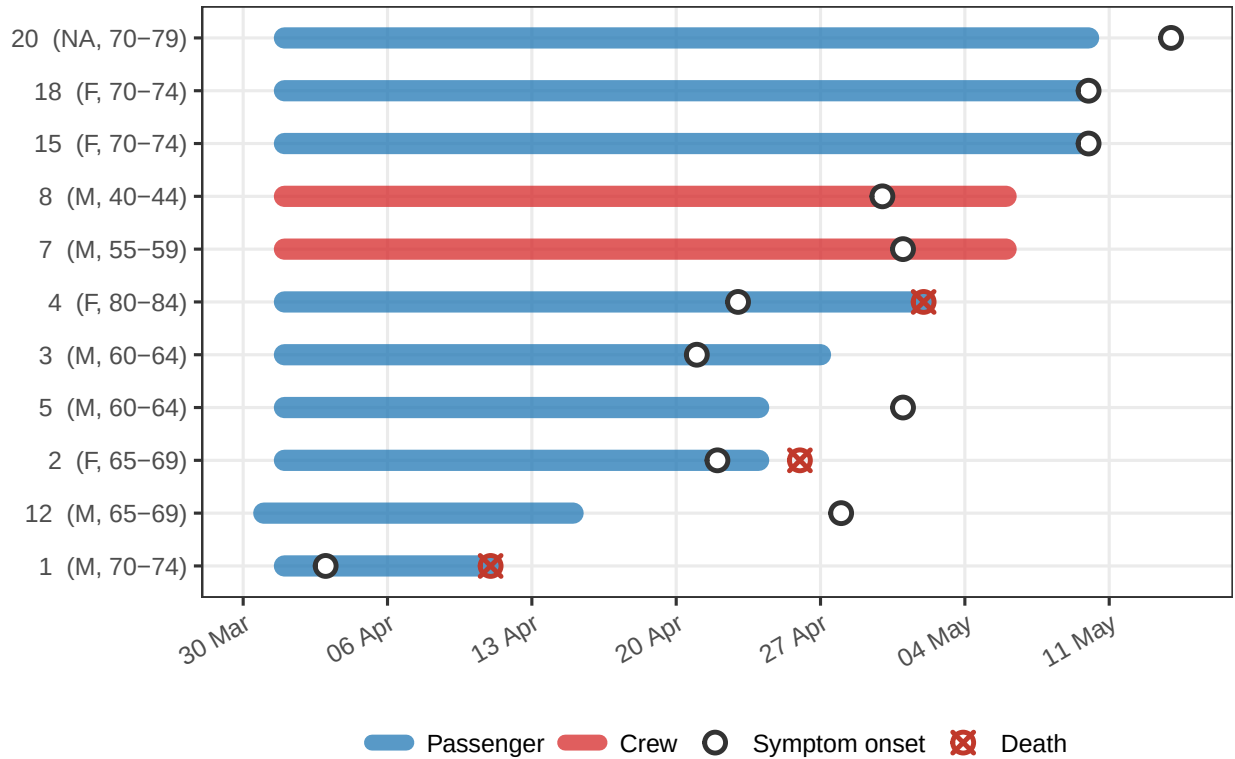


Figure 1: Ship exposure windows per case. Bars coloured by role; circle = symptom onset; circle-X = death.

Genomic Analysis

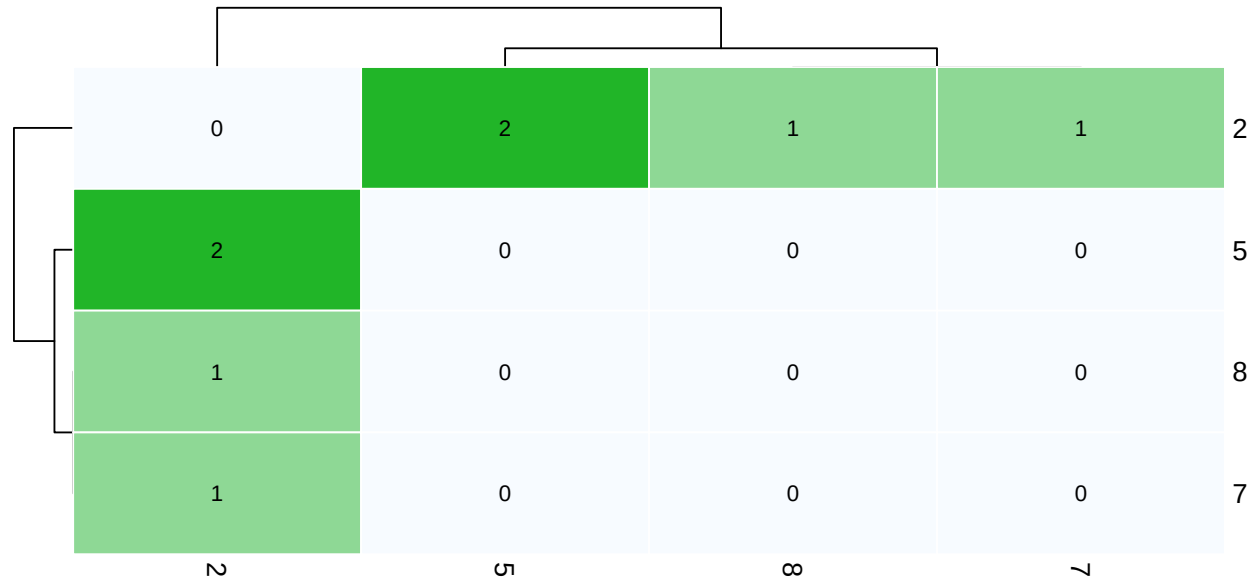


Figure 2: Pairwise SNP distance matrix for sequenced cases (L segment). Values are raw nucleotide differences. Near-zero distances confirm a near-clonal cluster consistent with a single spillover event.

Table 1: Convergence diagnostics across 4 independent chains. $R\text{-hat} < 1.01$

Parameter	R-hat	ESS	ESS: effective sample size out of 7200 total posterior samples. This row shows the worst-case (max R-hat, min ESS) across all parameters.
Reporting rate (.)	1.001	7,385	
Infection times (t_{inf} , $N = 13$)	1.002	3,164	

Model Diagnostics

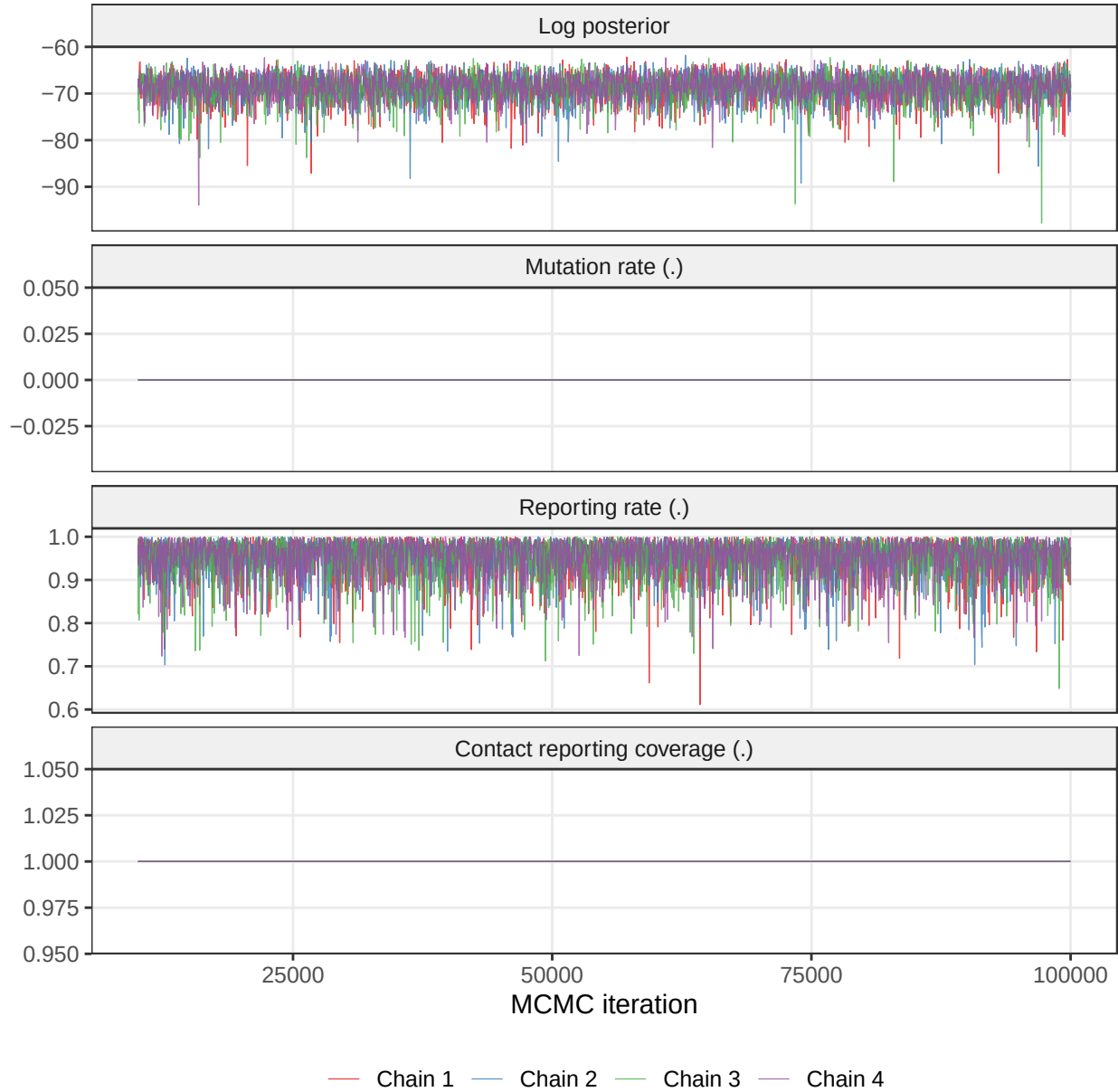


Figure 3: MCMC trace plots for scalar parameters (post-burnin). Stationary, well-mixed traces indicate convergence.

Transmission Tree

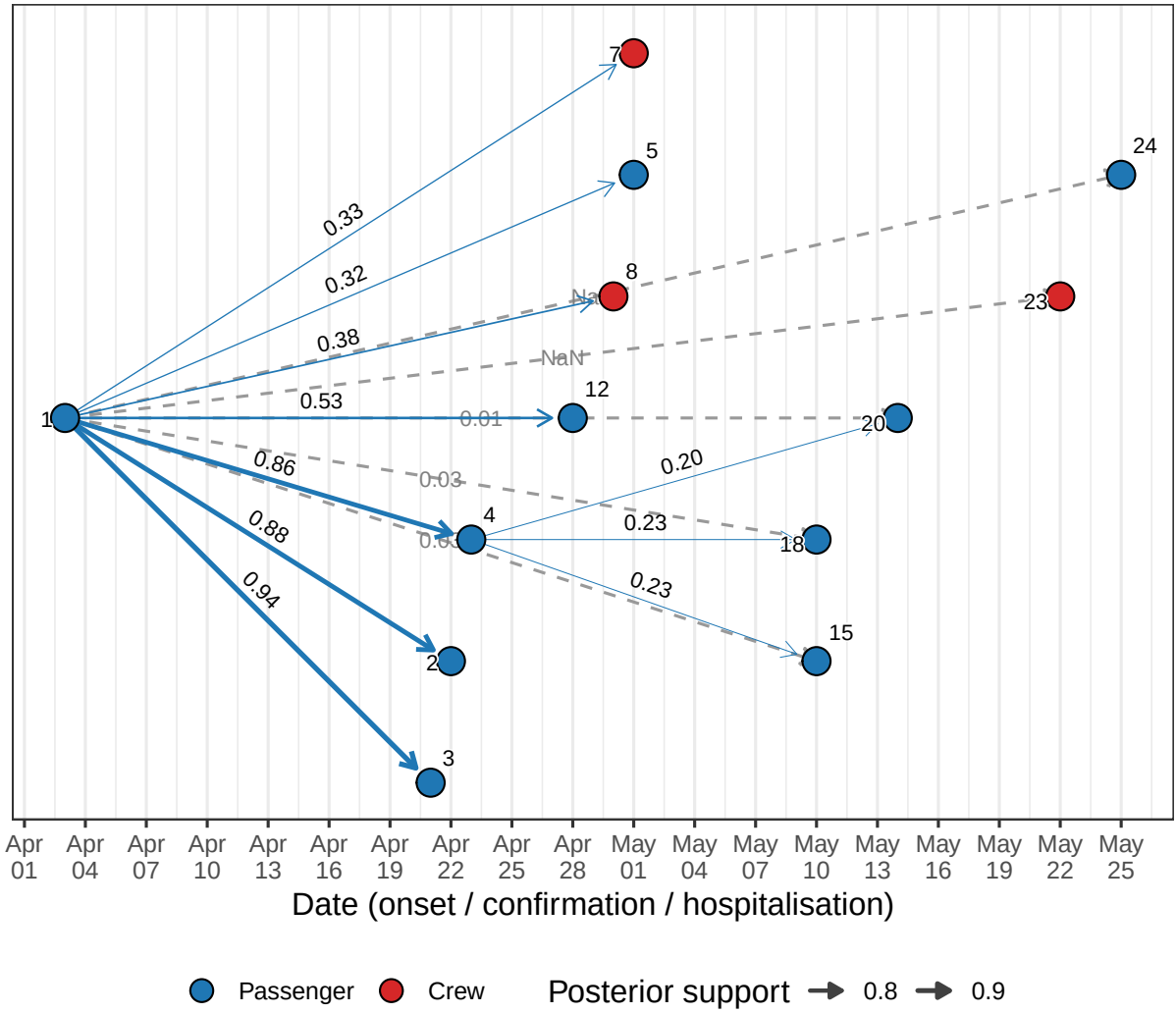


Figure 4: Consensus transmission tree (coloured edges), where each case is assigned its most frequently inferred ancestor across the posterior sample. Grey dashed arrows show the posterior support for direct transmission from case 1 to all cases not already linked to it in the consensus tree, consistent with a superspreading event.

Ancestry Uncertainty

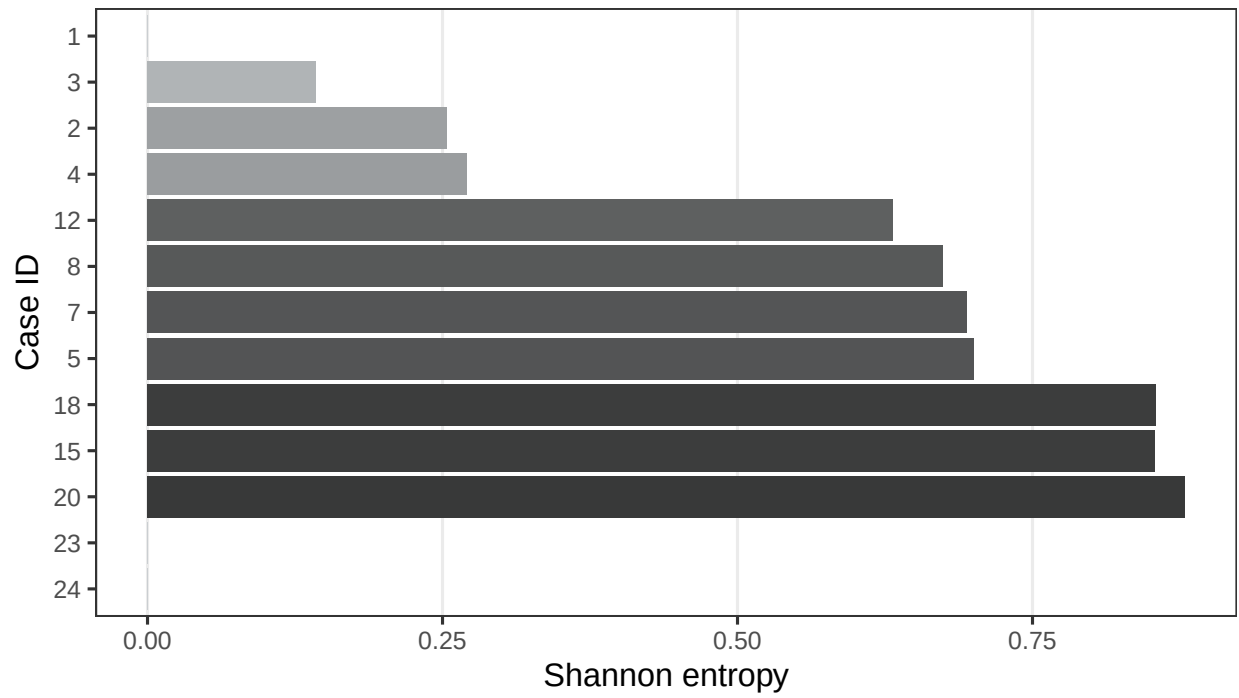


Figure 5: Shannon entropy of each case's inferred source. Entropy = 0 means a single unambiguous infector.

Dates of Infection



Figure 6: Posterior distribution of inferred infection dates per case. Width reflects uncertainty; cases ordered by mean inferred infection date.

Imports and Unobserved Transmission

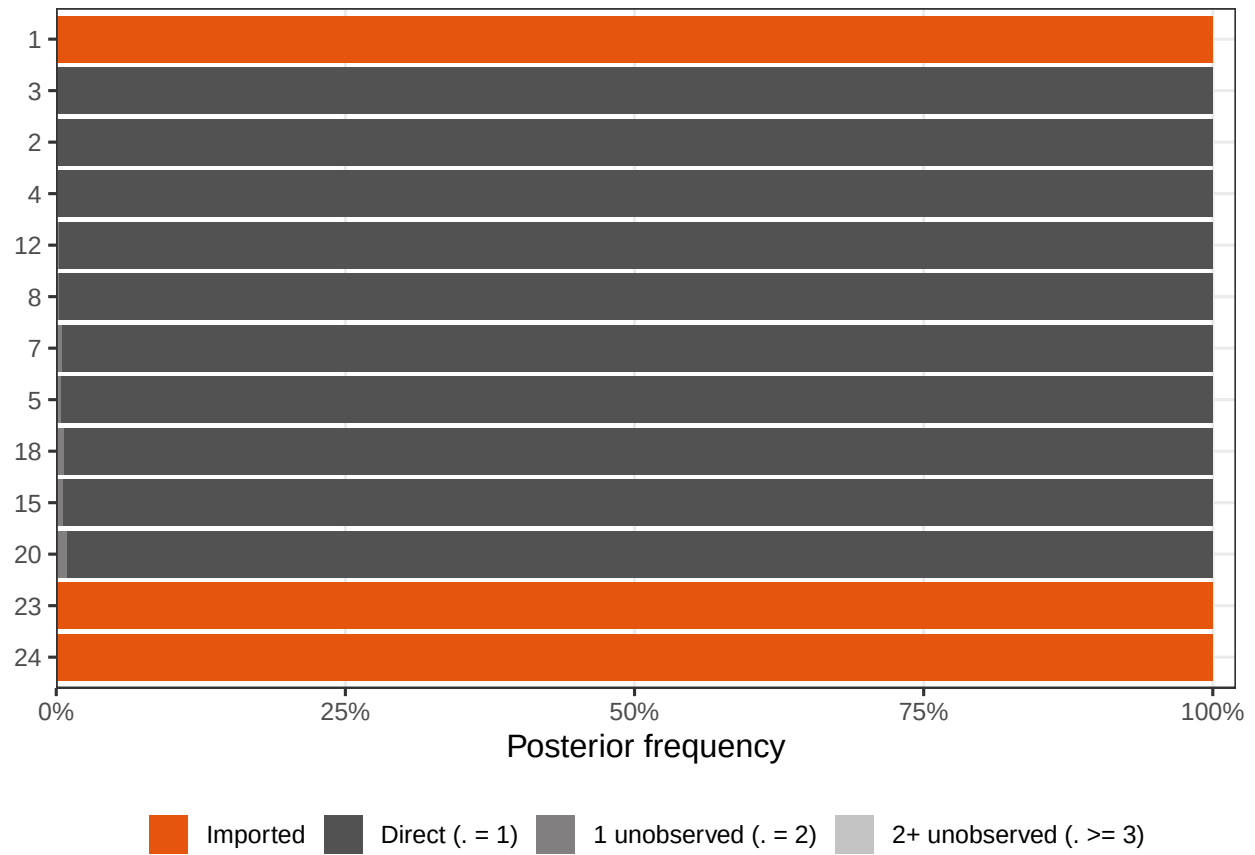


Figure 7: Posterior probability of each case being imported, directly linked, or connected via unobserved intermediate cases ().

Reproduction Number

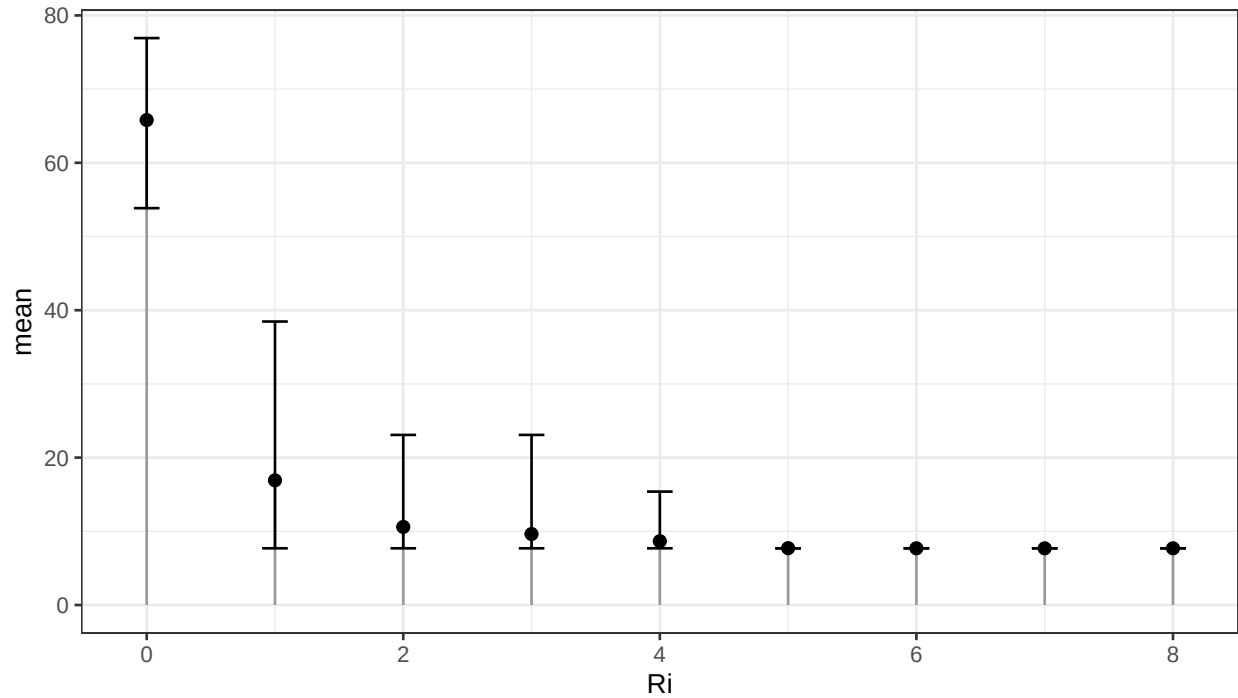


Figure 8: Posterior distribution of the offspring distribution.

Note

Cases 23 (WHO Case 12, Dutch crew member) and 24 (WHO Case 13, Spanish national) are inferred as imported by the model. Both lack symptom onset dates — the model falls back to their confirmation dates (22 and 25 May), placing them 49 and 52 days after the index case. At those dates, the serial interval likelihood of linkage to any earlier case is very low. Neither case has a sequenced sample, removing the genetic signal that would otherwise support a specific ancestry. Their ship disembarkation dates are also absent from the linelist, so the shared-location contact window contributes no additional evidence. Combined with the aggressive import threshold (`outlier_threshold = 2.5`), the model consistently assigns `= NA` to both cases. They may well be secondary cases within the ship chain; the classification would likely change if onset dates or sequences become available.

Data

Data available under [CC BY 4.0](#) at [kraemer-lab/Hondius_hantavirus_h2026](#). Sequences retrieved from [Pathoplexus](#). Report source: `report.qmd`.